

Regional Quantitative Problem-Solving Circle

Solution Key 3: Number Theory & Modular Congruence (Weeks 5–6)

Lead Architect: Mohamed Jad Srifi

Term: Fall 2025 – Spring 2026

Rigorous Proofs & Structural Solutions

Problem 1 (Polynomial Divisibility & Euclidean Reduction — AIME Caliber). Find all positive integers n such that $(2n + 1) \mid (n^3 + 3n^2 + 7)$.

Proof. Let $A = 2n + 1$ and $B = n^3 + 3n^2 + 7$. We are given the condition $A \mid B$. The fundamental invariant of Euclidean divisibility states that if $A \mid B$, then A must also divide any linear combination involving B , such as $A \mid 8B$. Multiplying the target polynomial by 8 allows us to cleanly execute polynomial long division without introducing fractional coefficients:

$$8B = 8n^3 + 24n^2 + 56$$

We divide $8n^3 + 24n^2 + 56$ by $2n + 1$ to isolate a constant remainder:

$$8n^3 + 24n^2 + 56 = (2n + 1)(4n^2 + 10n - 5) + 61$$

Because $(2n + 1)$ trivially divides the component $(2n + 1)(4n^2 + 10n - 5)$, the structural invariant of divisibility dictates that it must also divide the isolated remainder:

$$(2n + 1) \mid 61$$

Since 61 is a prime number, its only integer divisors are $-61, -1, 1, 61$. We test each divisor boundary:

- $2n + 1 = 1 \implies 2n = 0 \implies n = 0$ (Not a positive integer).
- $2n + 1 = -1 \implies 2n = -2 \implies n = -1$ (Not a positive integer).
- $2n + 1 = -61 \implies 2n = -62 \implies n = -31$ (Not a positive integer).
- $2n + 1 = 61 \implies 2n = 60 \implies n = 30$.

Thus, the only positive integer solution is $n = 30$. □

Problem 2 (Order, Invariants & Fermat's Little Theorem — Olympiad Track). Let p be an odd prime. Prove that any prime divisor q of the Mersenne number $2^p - 1$ must satisfy the congruence $q \equiv 1 \pmod{2p}$.

Proof. Since q is a prime divisor of $2^p - 1$, we can express this relation using modular congruence:

$$2^p - 1 \equiv 0 \pmod{q} \implies 2^p \equiv 1 \pmod{q}$$

This congruence implies that the multiplicative order of 2 modulo q , denoted $\text{ord}_q(2)$, must divide the exponent p . Because p is prime, its only positive divisors are 1 and p . If $\text{ord}_q(2) = 1$, this would mean $2^1 \equiv 1 \pmod{q}$, implying $q \mid 1$, which is impossible for a prime q . Therefore, the multiplicative order is strictly determined: $\text{ord}_q(2) = p$.

By Fermat's Little Theorem, since q is an odd prime (as it divides an odd number) and $\gcd(2, q) = 1$, we know that:

$$2^{q-1} \equiv 1 \pmod{q}$$

A fundamental property of multiplicative orders is that if $2^k \equiv 1 \pmod{q}$, then $\text{ord}_q(2) \mid k$. Substituting our known order, we deduce:

$$p \mid (q - 1)$$

This means there exists an integer k such that $q - 1 = kp$. Now we invoke parity. The number $2^p - 1$ is strictly odd. Any prime divisor q of an odd number must also be odd. Consequently, the term $q - 1$ must be an even integer. Since p is an odd prime, for the product kp to be even, the integer multiplier k must be even. Let $k = 2m$. Substituting this back into the equality yields:

$$q - 1 = (2m)p = m(2p)$$

This factorization proves that $2p \mid (q - 1)$, which is definitionally equivalent to the congruence:

$$q \equiv 1 \pmod{2p}$$

□

Problem 3 (Diophantine Integral Invariant — MathMaroc / Putnam Style). Determine all pairs of positive integers (x, y) satisfying the equation $x^2 - 3y^2 = 17$.

Proof. Assume, for the sake of contradiction, that there exists at least one pair of positive integers (x, y) satisfying the equation. We analyze the Diophantine equation by projecting it into the modular ring $\mathbb{Z}/3\mathbb{Z}$ (modulo 3). Taking the entire equation modulo 3, the term $-3y^2$ is perfectly divisible by 3 and evaluates to 0:

$$x^2 - 0 \equiv 17 \pmod{3}$$

Reducing the constant 17 modulo 3, we find $17 = 3 \times 5 + 2$, so $17 \equiv 2 \pmod{3}$. This leaves us with the constrained congruence:

$$x^2 \equiv 2 \pmod{3}$$

We now evaluate the exhaustive set of quadratic residues modulo 3. Any integer x must be congruent to 0, 1, or 2 modulo 3. Squaring these base residue classes:

- If $x \equiv 0 \pmod{3}$, then $x^2 \equiv 0 \pmod{3}$.
- If $x \equiv 1 \pmod{3}$, then $x^2 \equiv 1 \pmod{3}$.
- If $x \equiv 2 \pmod{3}$, then $x^2 \equiv 4 \equiv 1 \pmod{3}$.

The only valid quadratic residues modulo 3 are 0 and 1. The value 2 is a strict quadratic non-residue. Because no integer x can satisfy $x^2 \equiv 2 \pmod{3}$, the modular constraint contradicts the original assumption. Thus, the equation $x^2 - 3y^2 = 17$ admits strictly no integer solutions. □